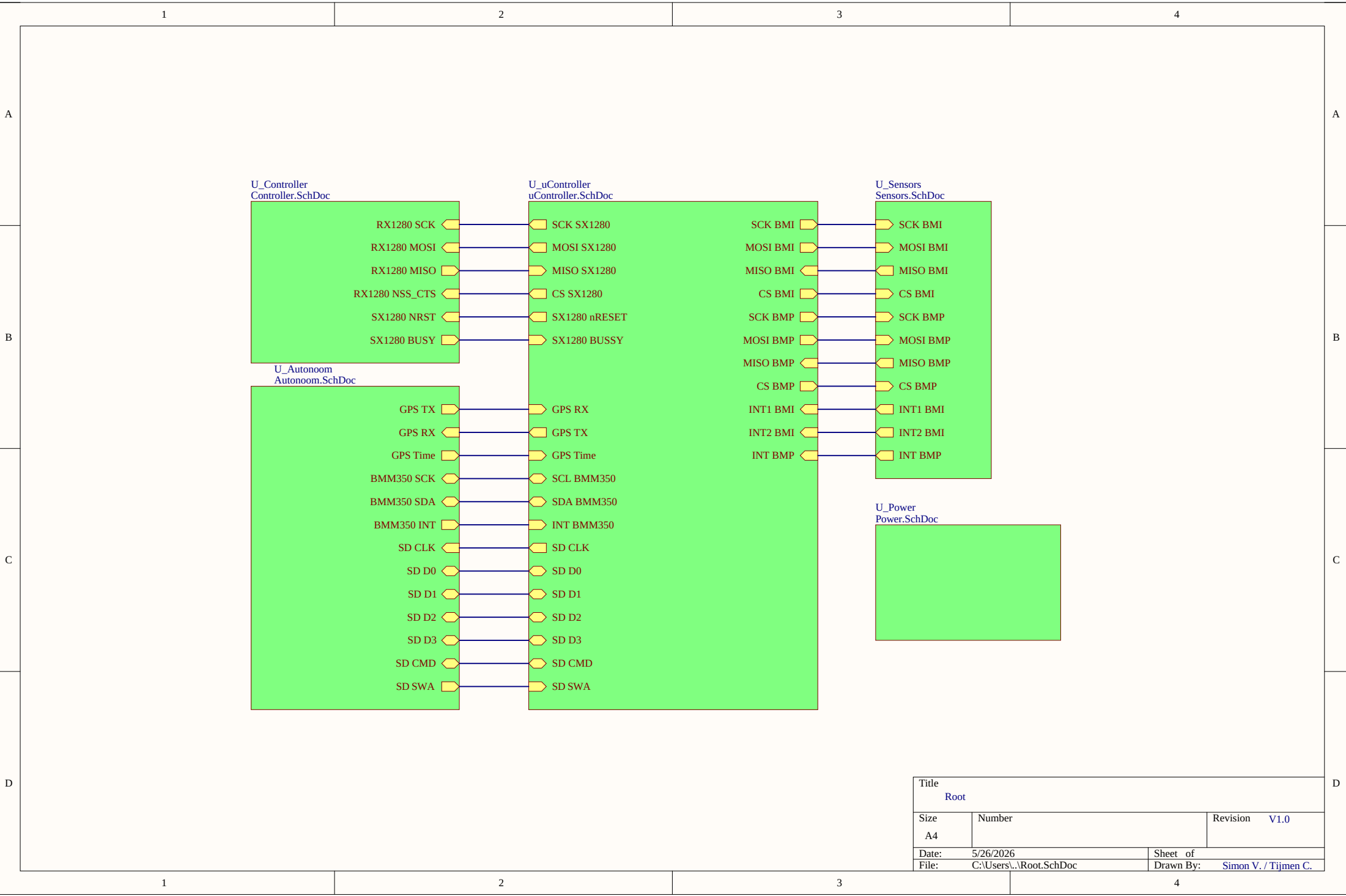
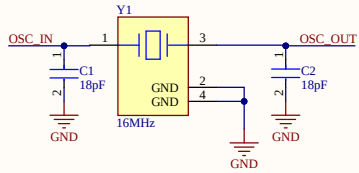




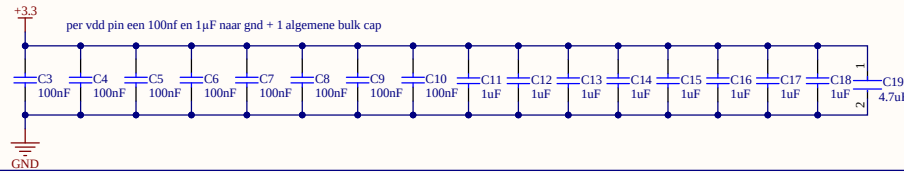
Title			
Root			
Size	Number		Revision
A4			V1.0
Date:	5/26/2026	Sheet of	
File:	C:\Users\...\Root.SchDoc	Drawn By:	Simon V. / Tijmen C.

Clock



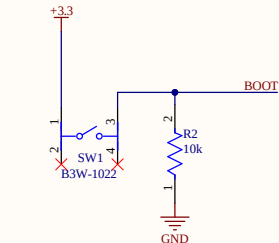
*voor layout: zo dicht mogelijk bij stm

POWER Filtering



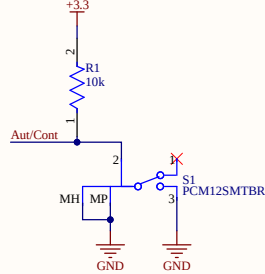
VCAP1 en 2 zijn voor core voltage => moet heel stabiel zijn vandaar de caps
BYPASS-reg = naar gnd zet de interne core spanningsregelaar aan
PDR_ON: Power Down Reset = sluit mooi af bij power uit, zet hoog om in te schakelen

BOOT

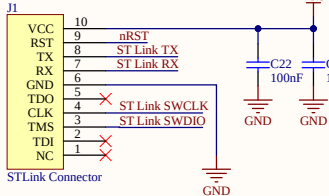


BOOT0 LOW = RUN, HIGH = Program
SCHAKELAAR NIET VOORZIEN
Maak soldeer pad voor moest ST link niet werken

Autonom / Controller

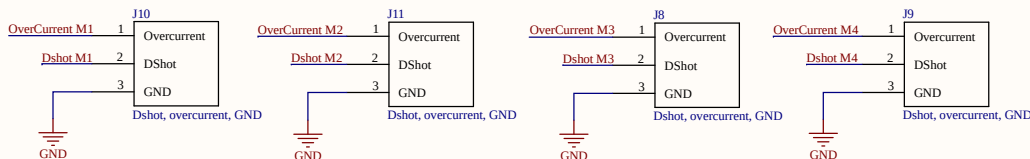


STLINK

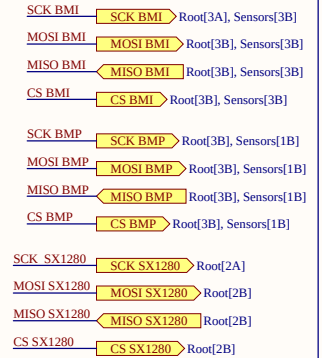


STLink Connector

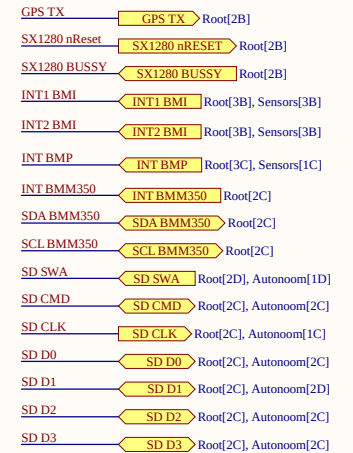
Flight Controller To ESC



SPI



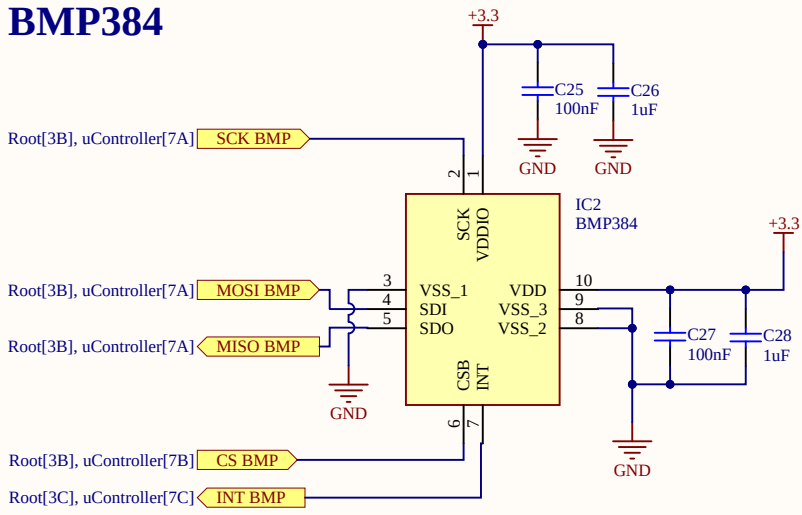
GPS Timer -> GPS Time -> Root[2C], Autonom[3C]
Naar PA0 want 32 bit TIMER Interrupt
GPS RX -> GPS RX -> Root[2B]



Title		
Size A3	Number	Revision V1.0
Date: 5/26/2026	Sheet of	
File: C:\Users\...uController.SchDoc	Drawn By: Simon V. / Tijmen C.	

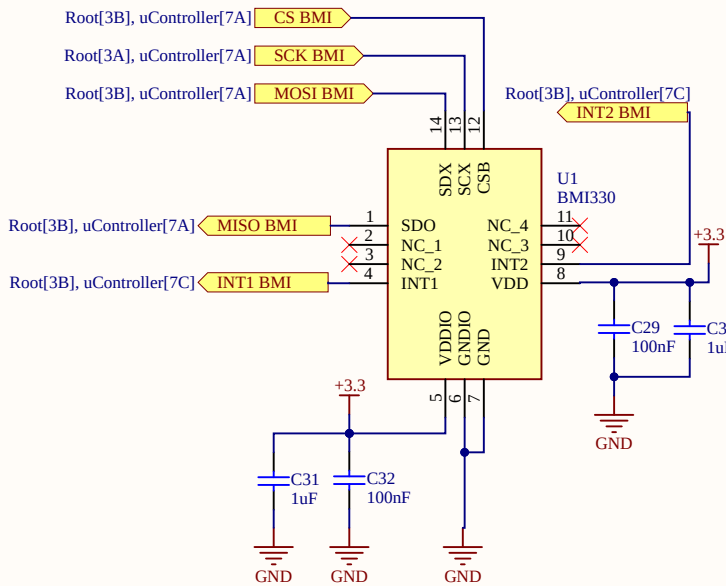
De Interrupt pin wordt niet gebruikt want hoogte verandert
relatief traag dus µcont. moet niet direct weten wnr er nieuwe
data is. in plaats daarvan gaan we hier polling gebruiken

BMP384



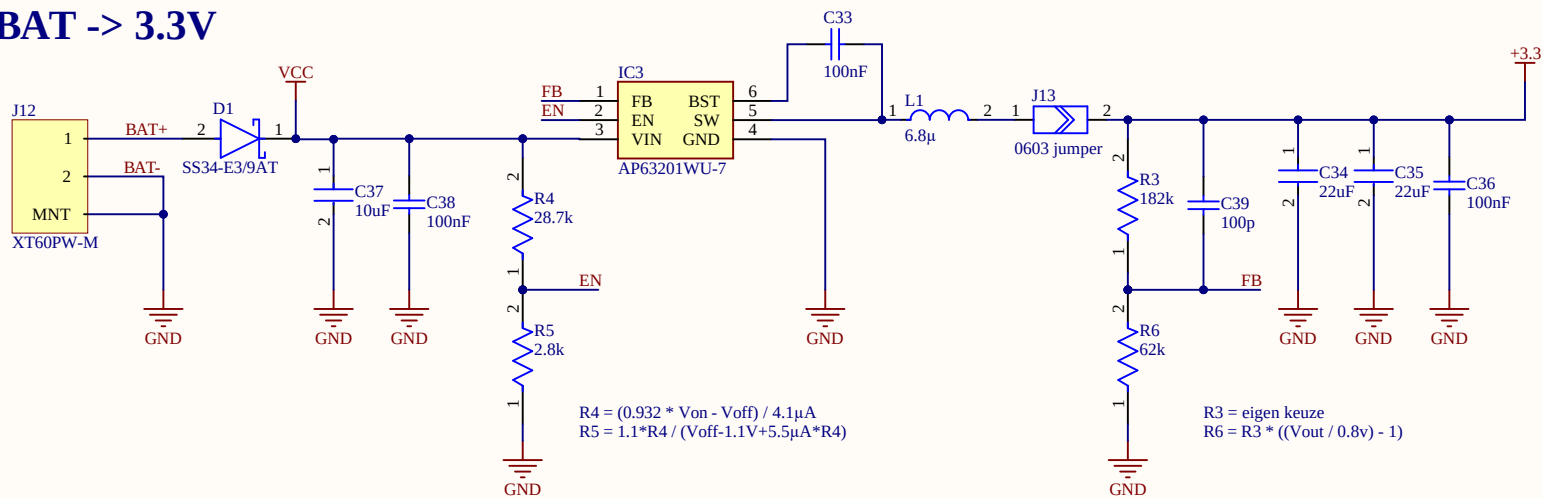
BMI330

ASDX en ASCX zijn voor I2C OSDO en OCSB zijn voor OIS control unit

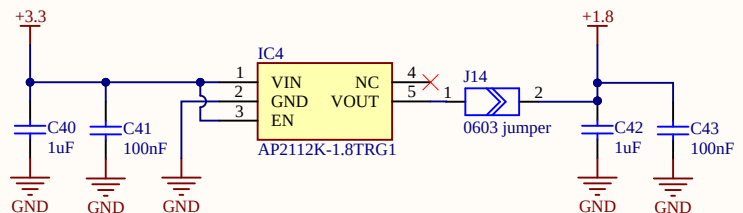


Title Sensors			
Size A4	Number		Revision V1.0
Date:	5/26/2026	Sheet of	
File:	C:\Users\...\Sensors.SchDoc	Drawn By:	Simon V. / Tijmen C.

BAT -> 3.3V

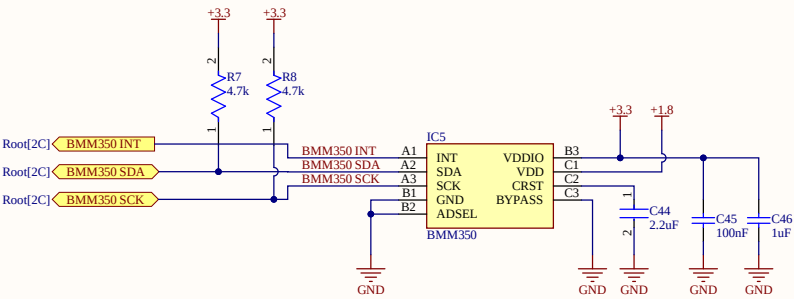


3.3V -> 1.8V

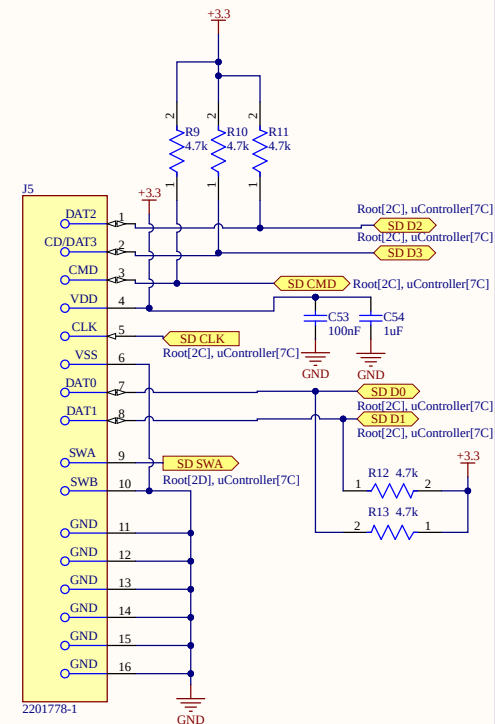


Title			
Power			
Size	Number	Revision	
A4		V1.0	
Date:	5/26/2026	Sheet of	
File:	C:\Users\...\Power.SchDoc	Drawn By:	Simon V. / Tijmen C.

Condensator op CRST is een specifieke condensator

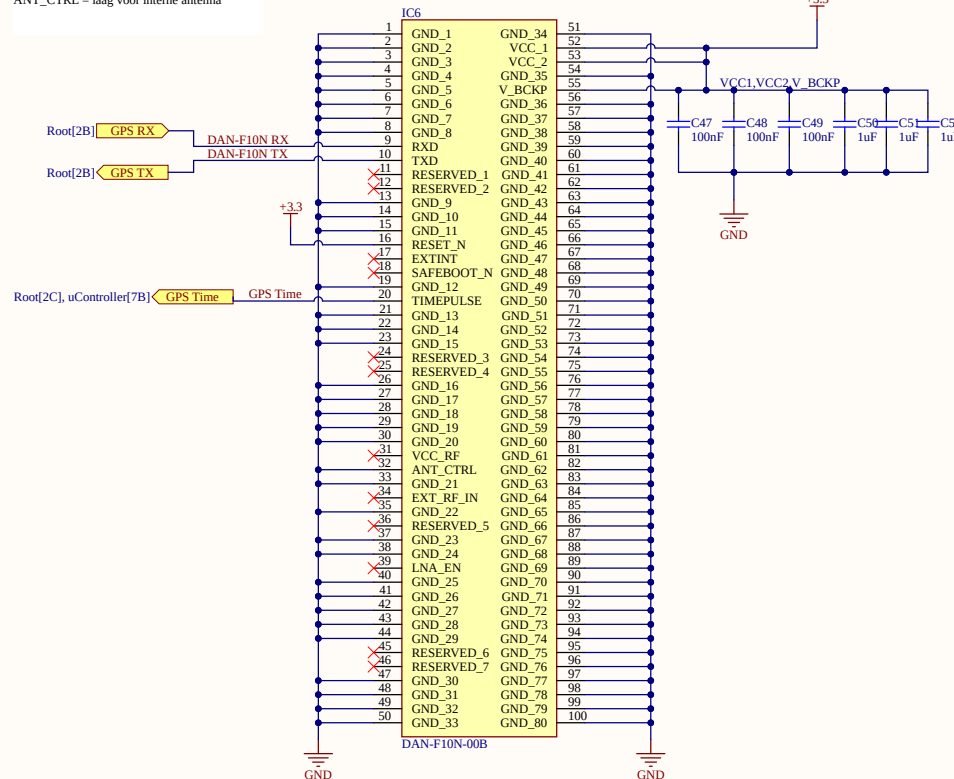


SDMMC interface
Software FATFS

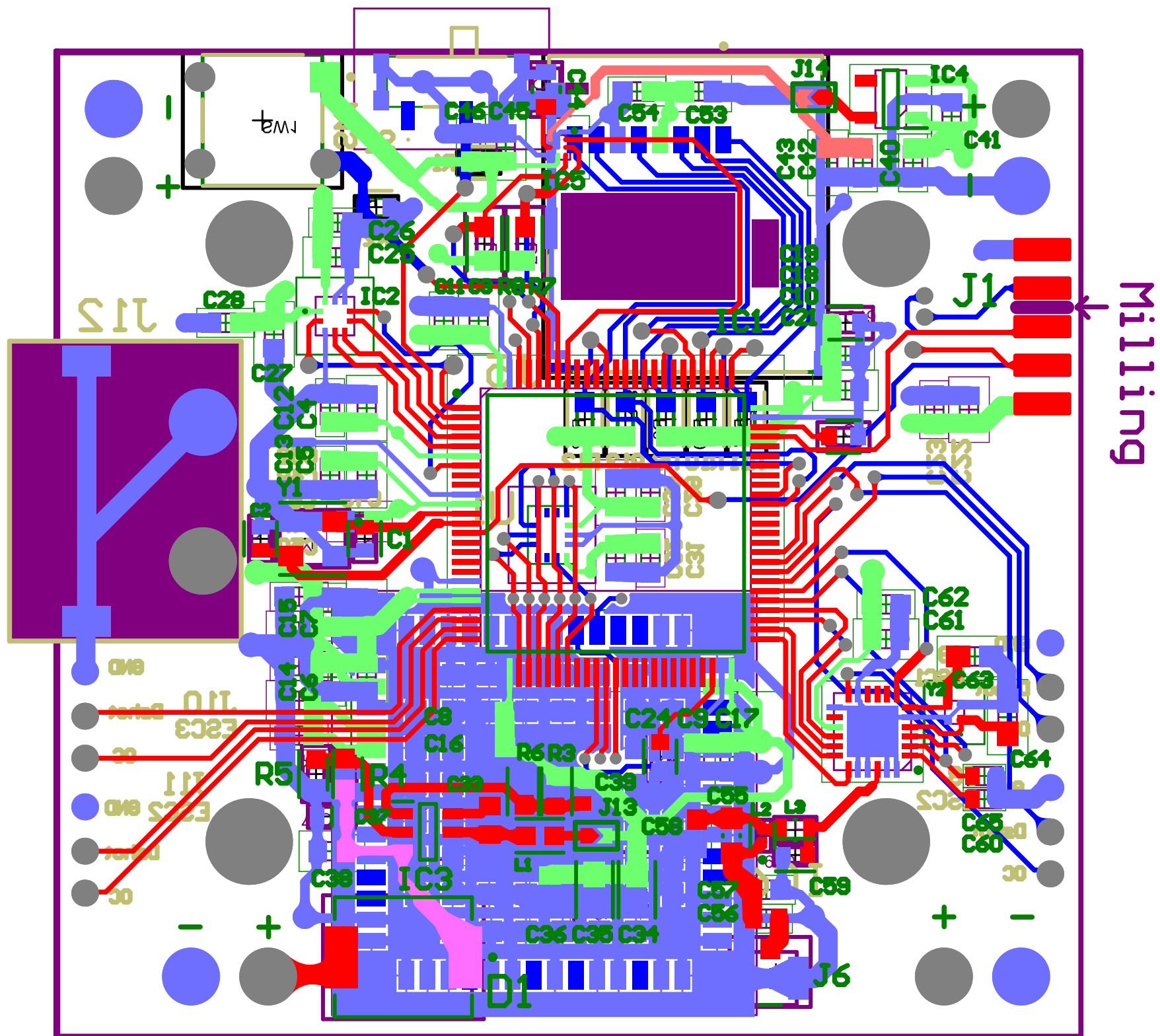


GPS + AntennA

ANT_CTRL = laag voor interne antenna



Title Autonoom			
Size A3	Number		Revision V1.0
Date: 5/26/2026	Sheet of		
File: C:\Users\...\Autonoom.SchDoc	Drawn By: Simon V. / Tijmen C.		



Line #	Name	Description	Designator	Revision ID	Revision State	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	18pF	C0603 18 pF CGO 30ppm/°C 1.00% 50 V	C1, C2			Unknown server		2 Yageo Group	C0603C180F5GACAUTO	Volume Production				
	100nF		C3, C4, C5, C6, C7, C8, C9, C10, C22, C25, C27, C29, C32, C33, C36, C38, C41, C43, C45, C47, C48, C49, C53, C61			Unknown server	24	Vishay	VJ0603Y104MXXAC	Volume Production				
	1uF		C11, C12, C13, C14, C15, C16, C17, C18, C23, C26, C28, C30, C31, C40, C42, C46, C50, C51, C62, C64, C62			Unknown server	21	KYOCERA AVX	0603YD105MAT2A	Volume Production				
	4.7uF	CAP CER 4.7UF 6.3V X7R 0603	C19			Unknown server	1	Yageo Group	CC0603MRX7R5B8475	Volume Production				
	2.2uF	CC0603 2.2 uF XSR 10.00% 16 V	C21, C24			Unknown server	2	Yageo Group	CC0603KFX5R7B8225	Volume Production				
	22uF	Multilayer Ceramic Capacitor 22uF 20V 10V XSR 0603 Paper T/R	C34, C35			Unknown server	2	Murata	GRM188R61C226ME01D	Volume Production				
	10uF	CAP CER 10UF 25V XSR 0603	C37			Unknown server	1	Murata	ZR818AR61E106ME01L	Volume Production				
	100p		C39			Unknown server	1	Vishay	VJ0603A101FXAAC	Volume Production				
	2.2uF	Multilayer Ceramic Capacitors 2.2uF ±10% 6.3V X7R SMD 0605, type: CGB4B3X7R0J225K 055AB	C44			Unknown server	1	TDK	CGB4B3X7R0J225K055AB	End of Life				
	0.5pF	CAP 0.5pF 10V ±0.1pF 0603 (1608 Metric) Thickness 1mm SMD	C55			Unknown server	1	Vishay	VJ0603A0R5BXCQW1BC	Volume Production				
	100pF		C56			Unknown server	1	Vishay	VJ0603A101KXCAC	Volume Production				
	1.5pF		C57, C58			Unknown server	2	KYOCERA AVX	0603A1R2CAT2A	Volume Production				
	0.8pF	Ceramic Capacitor for Automotive 0.8pF ±0.1pF 100VDC CGO 0603 Paper T/R	C59			Unknown server	1	Murata	GCM1885CA8R08A16J	Volume Production				
	10nF		C60, C65			Unknown server	2	Vishay	VJ0603Y103JXXAC	Volume Production				
	9pF		C63, C64			Unknown server	2	KYOCERA AVX	0603LA5R0CAT2A	Volume Production				
	SS34-E3/9AT	DIODE SCHOTTKY 40V 3A D09234AB	D1			Unknown server	1	Vishay	SS34-E3/9AT	Volume Production				
	STM32F745VGT6	Integrated Circuit	I1			Not managed	1							
	8MP384	Integrated Circuit	I2			Not managed	1							
	AP63201WL-7	Integrated Circuit	I3			Not managed	1							
	AP7112K-1.8TRG1	Integrated Circuit	I4			Not managed	1							
	BMM350	Integrated Circuit	I5			Not managed	1							
	CAH-F120V-009	Integrated Circuit	I6			Not managed	1							
	SX1280MLTRT	Integrated Circuit	I7			Not managed	1							
	STLink Connector		J1			Not managed	1							
	2201778-1	MICRO 3D PUSH PUSH LOW PROFILE T	J5			Unknown server	1	TE Connectivity	2201778-1	Volume Production				
	UJFL-R-SMT(10)	CONN ULFL RCPT STR 50 OHM SMD	J6			Unknown server	1	HIROSE ELECTRIC	UJFL-R-SMT(10)	Volume Production				
	Dsphot, overcurrent, CND		J8, J9, J10, J11			Not managed	4							
	XT60PW-M		J12			Not managed	1							
	0603 jumper		J13, J14			Not managed	2							
	6.8u	Shielded Multilayer Inductor 6.8uH ±20% 300mA 962mq SMD 0603	L1			Unknown server	1	TDK	ML21608MR6RW7000	Volume Production				
	2.5nH	Film type RF Inductor 2.5nH ±0.1nH 500mA 0.20 0201 (0603)	L2			Unknown server	1	Murata	LQP03TN2N5B02D	Volume Production				
	3nH	Wire Wound RF Inductor 3nH ±0.2nH 670mA 0.170 0603 (1608)	L3			Unknown server	1	Murata	LQW18AN3NOC80D	Volume Production				
	10k	RES SMD 100 OHM 1% 1/8W 0603	R1, R2			Unknown server	2	Vishay	MCT0603HE1002BP100	Volume Production				
	162k	RES 162kΩ 1.00% 0.1W 0603	R3			Unknown server	1	Yageo Group	AC0603FR-07162KL	Volume Production				
	28.7k	RES Thin Film, 28.7kΩ, 0.1%, 0.1W, ±25ppm/°C, 0603	R4			Unknown server	1	Venkel	TFCR0603-10W-E-2872BT-1K	Volume Production				
	2.8k	RES Thick Film, 2.8kΩ, 1%, 0.33W, 100ppm/°C, 0603	R5			Unknown server	1	Vishay	CRCW06032K80FKEAHP	Volume Production				
	62k	62kΩ ±5% 0.1W 0603 Thick Film Chip Resistor AEC- Q200 compliant	R6			Unknown server	1	Stackpole Electronics	RMCF0603T62K0	Volume Production				
	4.7k	RES 4.7kΩ 1.00% 0.1W 0603	R7, R8, R9, R10, R11, R12, R13			Unknown server	7	Yageo Group	AC0603FR-104K7L	Volume Production				
	FCM125MTBR		S1			Not managed	1							
	B3W-1022	SWITCH TACTILE SPST-NO 0.05A 24V	SW1			Unknown server	1	Omron	B3W-1022	Volume Production				
	BM1330	Undefined or Miscellaneous	U1			Not managed	1							
	16MHz	Crystal 16MHz ±20ppm 18pF SMD-4 3.2mm x 2.5mm ABM6-16.000MHZ- B2-T3	Y1			Unknown server	1	Abracon	ABM6-16.000MHZ-B2-T3	Volume Production				
	ABM13W- 52.000MHZ-5-B2G- T5	Crystal or Oscillator	Y2			Not managed	1							